

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper: CSB-504 — Operating System Concepts

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all, including Question No. 1, which is compulsory. Figures on the right-hand side indicate full marks.

Question 1. Answer all parts: [3 + 4 + 4 + 5 + 6 = 22]

- (a) Briefly explain the main design goals of an operating system.
- (b) What are the differences between a process and a thread? Describe some benefits of using threads.
- (c) Explain the difference between internal and external fragmentation. Why should they be avoided?
- (d) List the four conditions for the occurrence of a deadlock and describe the resource-allocation graph.
- (e) Briefly discuss the following terms: Dynamic Loading, Busy Waiting, and Monitors.

Question 2.

- (a) Briefly explain the different process states and transitions among them with a state transition diagram. [2]
- (b) Explain the terms Process Control Block (PCB) and Context Switching. [4]
- (c) There are five processes (A to E) to run. Their arrival times are 0, 1, 3, 9, and 12 seconds, respectively. Their CPU burst times are 3, 5, 2, 5, and 5 seconds, respectively. What is the average turnaround time using First-Come-First-Served (FCFS), Shortest-Job-First (SJF), and Round-Robin (RR, with a 1-second time quantum) scheduling? [8]

Question 3.

- (a) Discuss the Paging memory management scheme with an illustrative example and explain the hardware support required for paging. [8]
- (b) Consider a paging system where the page table is stored in main memory. If a memory reference takes 200 nanoseconds, answer the following: [6]
 - (i) How long does a paged memory reference take?
 - (ii) If we add a TLB and 75% of all page-table references are found in the TLB, what is the effective memory reference time? Assume that a TLB lookup takes 20 nanoseconds (10 times faster than a memory access).

Question 4.

- (a) What is Demand Paging? Explain the steps involved in page fault handling. [4]
- (b) Briefly describe Belady's anomaly. [2]
- (c) A computer has 4 page frames. A process makes the following sequence of page references: 1, 2, 3, 4, 1, 5, 2, 3, 1, 2. How many page faults occur using FIFO, Second Chance, and Least-Recently Used (LRU) page replacement algorithms? [8]

Question 5.

- (a) Describe the physical and logical views of a disk storage system. [5]
- (b) Consider the following sequence of disk track requests: 27, 129, 110, 186, 147, 41, 10, 64, and 120. Assume that initially the head is at track 30 and is moving in the direction of decreasing track number. Compute the total head movement (in tracks) using FIFO, SSTF, and SCAN (Elevator) disk scheduling algorithms. [9]

Question 6.

- (a) Explain the contiguous and linked file allocation methods, and discuss the File Allocation Table (FAT) scheme for linked allocation. [6]
- (b) Explain the concepts involved in maintaining file system security. [4]
- (c) Consider a swapping system in which memory contains the following holes in memory order: 10 KB, 4 KB, 20 KB, 18 KB, 7 KB, 12 KB, and 15 KB. Which hole is allocated for successive segment requests of (i) 12 KB, (ii) 10 KB, and (iii) 9 KB, using First-fit, Best-fit, Worst-fit, and Next-fit allocation strategies? [4]

Question 7. Write short notes on any three of the following: [12]

- (a) Disk free-space management schemes
- (b) Types of Operating Systems
- (c) Semaphores
- (d) Deadlock Recovery